- It improves the water holding capacity and helps to prevent soil erosion.
- It contains valuable vitamins, enzymes and growth regulator substances for increasing growth, vigour and yield of plants.
- It enhances decomposition of organic matter in soil.
- Vermicompost is free from pathogens and toxic elements.
- Vermicompost is rich in beneficial microflora.

23.14 Apiculture

Apiculture is the rearing of honey bee for honey. It is also called Bee keeping. It is a profitable rural based industry. Honey bees are domesticated by farmers to produce honey.

23.14.1 Types of Honey Bee

There are three types of individuals in an honey bee colony namely the queen bee, the drones and the worker bees.

Queen Bee: The queen is the largest member and the fertile female of the colony. They are formed from fertile eggs. The queen is responsible for laying eggs in a colony.

Drones: They are the fertile males. They develop from unfertilized eggs. They are larger than the workers and smaller than the queens. Their main function is to fertilize the eggs produced by the queen.

Worker Bees: They are sterile female bees and are the smallest members of the colony. Their function is to collect honey, look after the young ones, clean the comb, defend the hive and maintain the temperature of the bee hive.

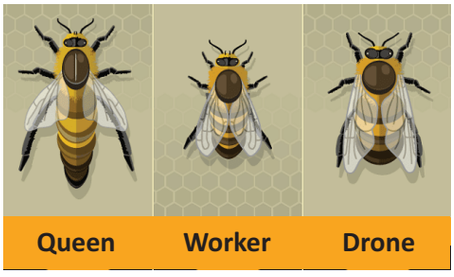


Figure 23.23 Types of Honey bee

23.14.2 Varieties of Honey Bee

Indigenous varieties

i) Apis dorsata (Rock bee or Wild bee)

ii) Apis florea (Little bee)

iii) *Apis indica* (Indian bee)

Exotic varieties

i) Apis mellifera (Italian bee)

ii) Apis adamsoni (African bee)

23.14.3 Structure of Bee Comb

The comb of the bees is formed mainly by the secretion of the wax glands present in the abdomen of the worker bee. A comb is a vertical sheet of wax with double layer of hexagonal cells.

Formation of Honey: The honey bees suck the nectar from various flowers. The nectar passes to the honey sac. In the honey sac, sucrose present in the nectar mixes with acidic secretion and by enzymatic action it is converted into honey which is stored in the special chambers of the hive.

Quality of honey depends upon the flowers available to the bees for nectar and pollen collection.

23.14.4 Products from Honey Bee

Honey bees are used in the production of honey and bee wax. Other useful products obtained from honey bees are bee pollen, royal jelly, propolis and bee venom.

Honey: Honey is a sweet, viscous, edible natural food product. Dextrose and sucrose gives sweet taste to the honey. It also contains amino acids, B-complex vitamins, ascorbic acid, and minerals Formic acid is a preservative in honey. Invertase is an enzyme present in honey.

Uses of Honey

- Honey has an antiseptic and antibacterial property. It is a blood purifier.
- It helps in building up of haemoglobin content in the blood.
- It is used in Ayurvedic and Unani system of medicines.
- It prevents cough, cold, fever and relieves sore throat.

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